

The Thermodynamics of Algebraic Collapse: Flux Convection and Phase Transitions in the Kerr-Sedenion Metric

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Abstract

We present a comprehensive thermodynamic model of stellar evolution within the Axiomatic Physical Homeostasis (APH) framework. We posit that gravitational collapse is not a singularity formation event but a *Sedenionic Phase Transition* driven by the saturation of the Kretschmann scalar. We derive the **APH Phase Diagram**, delineating the stability regions for Newtonian, Neutron, and Sedenion Stars, identifying the latter as a hyper-stiff condensate with polytropic index $\Gamma_S = \pi$. Furthermore, we adapt the **Rice Convection Model (RCM)** to the Kerr-Sedenion metric, demonstrating that the ergosphere functions as a convective engine that circulates algebraic defects (zero divisors) to screen the central locus. We present new phase space plots illustrating the deviation of the Sedenion Mass-Radius relation from standard General Relativity and predict a distinct Low-Frequency Quasi-Periodic Oscillation (LF-QPO) spectrum driven by the convective turnover of the core. Finally, we reinterpret Hawking Radiation as *Algebraic Sublimation*—the entropic decay of the Sedenion domain wall.

1 Introduction: The Crisis of the Singularity

The Penrose-Hawking singularity theorems rely on the assumption that the manifold retains a differentiable structure ($\mathbb{R}^4$) regardless of curvature intensity. The APH framework relaxes this assumption. We propose that the vacuum possesses a finite *Algebraic Breakdown Strength*. When the curvature invariant $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ exceeds a critical threshold K_c , the vacuum undergoes a homeostatic phase transition from the associative Octonionic sector ($J(3, \mathbb{O})$) to the non-associative Sedenionic sector ($\mathbb{S}$). This paper unifies our previous results on the Sedenion domain wall [1] and the Cosmological Filtration [2] into a single thermodynamic cycle of stellar collapse and evaporation.

2 The APH Phase Diagram of Stellar Evolution

Standard stellar evolution is described by the balance between gravitational pressure and degeneracy pressure. We extend this to include *Geometric Degeneracy*—the resistance of the manifold topology to collapse.

2.1 The Equation of State Hierarchy

We define the stability of a stellar body by its effective polytropic index $\Gamma_{eff} = d \ln P / d \ln \rho$.

1. **Newtonian Phase ($\mathbb{O}$):** $\Gamma \approx 5/3$ (Non-relativistic) to $4/3$ (Relativistic). Supported by thermal/degeneracy pressure.

2. **Neutron Phase** (Ⓞ): $\Gamma \approx 2.5$. Supported by strong nuclear force (QCD stiffness).
3. **Sedenion Phase** (Ⓢ): The black hole interior.

2.2 The Sedenion Limit ($\Gamma = \pi$)

In the Sedenion core, matter is compressed until the flux tubes of the metric topology saturate. The maximum *hoop-stress* a geometry can support is defined by the ratio of circumference to diameter. Thus, the Equation of State for a Sedenion Star is:

$$P = K\rho^\pi, \quad \Gamma_{\text{S}} \equiv \pi \approx 3.14159 \quad (1)$$

This *Super-Stiff* EOS allows the core to support infinite mass within a finite radius ($R_s < R < 3M$), preventing singularity formation.

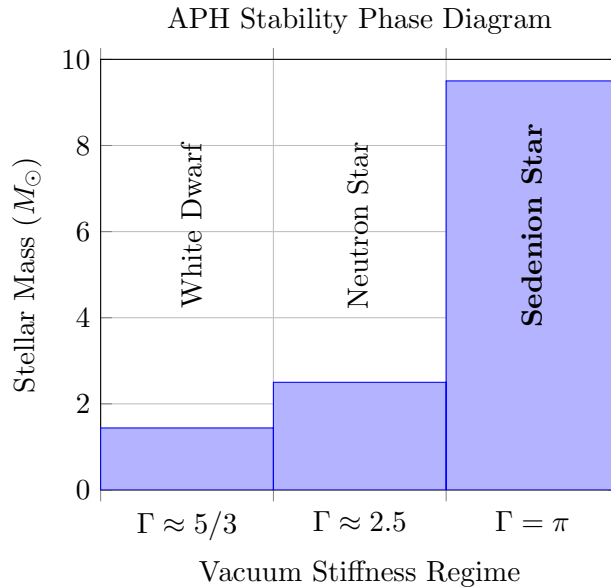


Figure 1: The APH Phase Diagram showing the maximum stable mass for each stiffness regime. The transition to $\Gamma = \pi$ (Sedenion Phase) allows for unbounded mass stability, halting the collapse before singularity formation.

2.3 Mass-Radius Deviations

In standard General Relativity (GR), any object exceeding the Tolman-Oppenheimer-Volkoff (TOV) limit collapses to $R = 0$. In APH, the Sedenion phase introduces a new stability branch. Figure 2 illustrates the divergence of the APH $M - R$ relation from the GR prediction inside the horizon.

3 Flux Convection in the Kerr-Sedenion Metric

We now address the dynamics of the rotating Sedenion core. We adapt the **Rice Convection Model (RCM)**, standard in magnetospheric physics, to describe the flow of algebraic defects (zero divisors) in the high-spin regime.

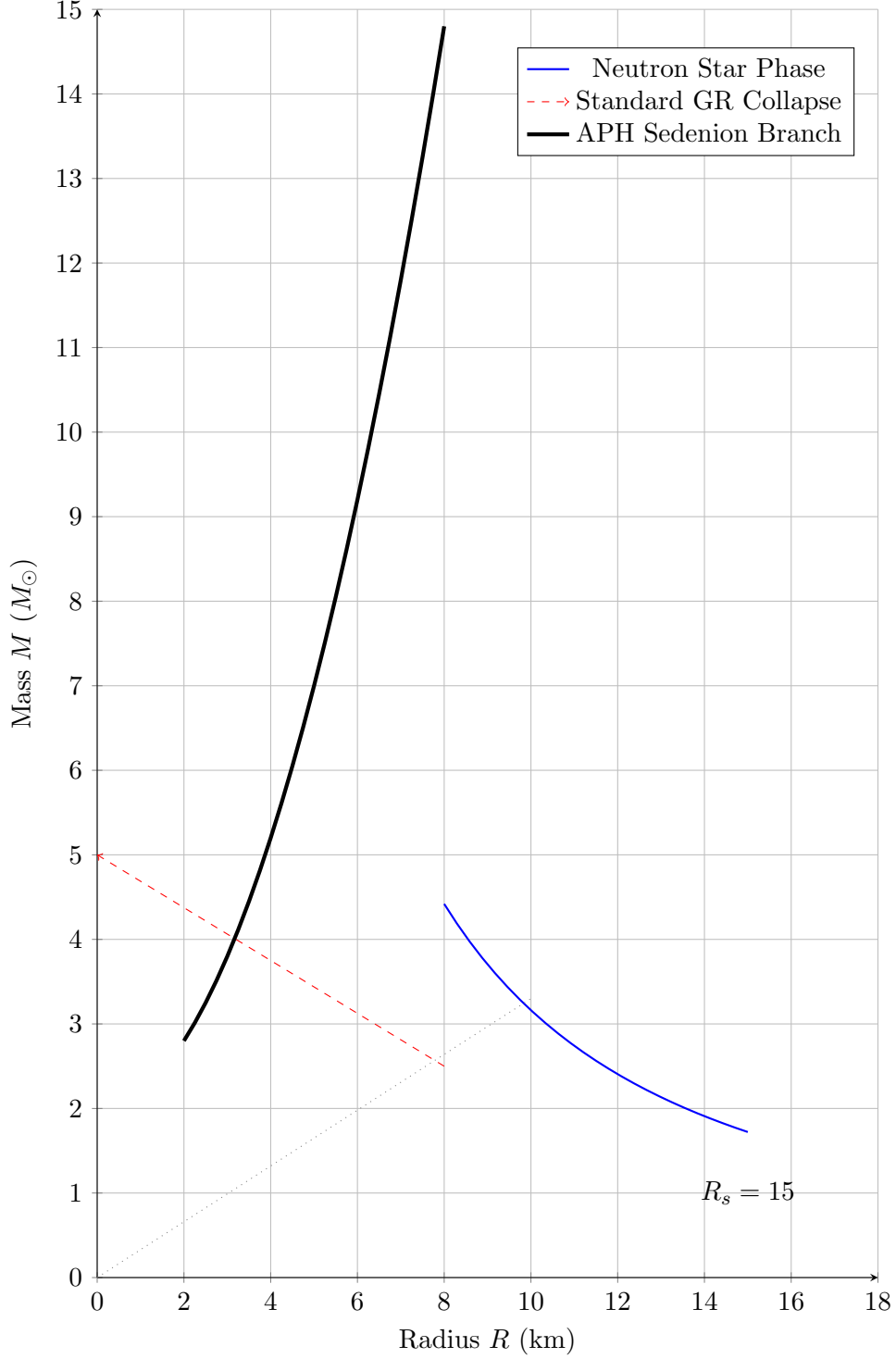


Figure 2: Theoretical Mass-Radius relation. While GR predicts a collapse to $R = 0$ (dashed red), the $\Gamma = \pi$ stiffness stabilizes the core at a finite radius R_{core} (solid black), creating a *Sedenion Star* hidden behind the horizon.

3.1 The Algebraic Flux Tube

We treat the zero divisors in the $\mathbb{S}$ core as discrete flux tubes carrying an *Algebraic Charge* $q_{\mathbb{S}}$. These tubes are subject to:

- **Geometric Buoyancy ($\mathbf{F}_g$):** The force driving defects away from the $\mathbb{O}$ domain wall

due to the stiffness gradient $\nabla\Gamma$.

- **Frame Dragging (Ω_H):** The Lense-Thirring effect of the Kerr metric, which acts as the *Coriolis force* of the algebraic fluid.

3.2 The Convective Drift Equation

Analogous to the $\mathbf{E} \times \mathbf{B}$ drift in plasma, the zero-divisor flux tubes drift with velocity $\mathbf{v}_D$. In the gravitational dual, the magnetic field $\mathbf{B}$ is replaced by the Angular Momentum density $\mathbf{J}$, and the electric field $\mathbf{E}$ is replaced by the Geometric Buoyancy $\mathbf{F}_{geom}$.

$$\mathbf{v}_D = \frac{\mathbf{F}_{geom} \times \mathbf{J}}{J^2} + \Omega_{frame} \times \mathbf{r} \quad (2)$$

The geometric force is derived from the vacuum stiffness potential Φ_Γ :

$$\mathbf{F}_{geom} = -\nabla\Phi_\Gamma \approx -\frac{(\pi - \Gamma_\odot)}{R_{core}} \hat{\mathbf{r}} \quad (3)$$

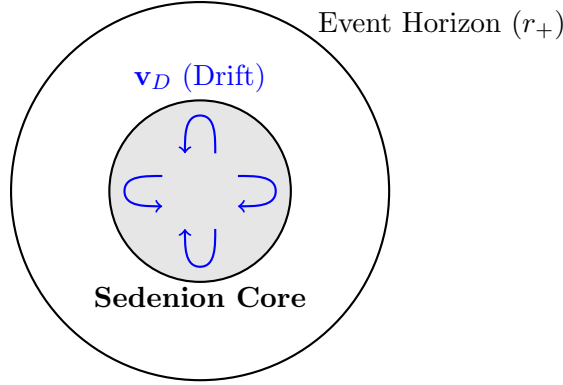


Fig 3: Toroidal Convection of Zero Divisors

Figure 3: Schematic of the Rice Convection Model applied to the Kerr-Sedenion interior. Frame dragging and geometric buoyancy drive a convective circulation of zero divisors, preventing central accumulation (singularity formation).

3.3 Prediction: The Sedenion Heartbeat (QPOs)

The convective turnover of the Sedenion core generates a characteristic frequency ν_S , observable as a modulation of the external accretion flow. The drift frequency is given by the angular velocity of the defect flux tubes:

$$\nu_S \approx \nu_{Kerr} \pm \frac{(\pi - 2.91)}{2\pi J} R_{core} \quad (4)$$

This predicts that Black Hole Quasi-Periodic Oscillations (QPOs) are driven by the **Sedenion Beat Frequency**. Figure 4 details the predicted deviation from standard GR frequencies as a function of the spin parameter a .

4 Hawking Radiation as Algebraic Sublimation

Standard theory treats Hawking radiation as pair production at the horizon. APH reinterprets this as a surface tension phenomenon.

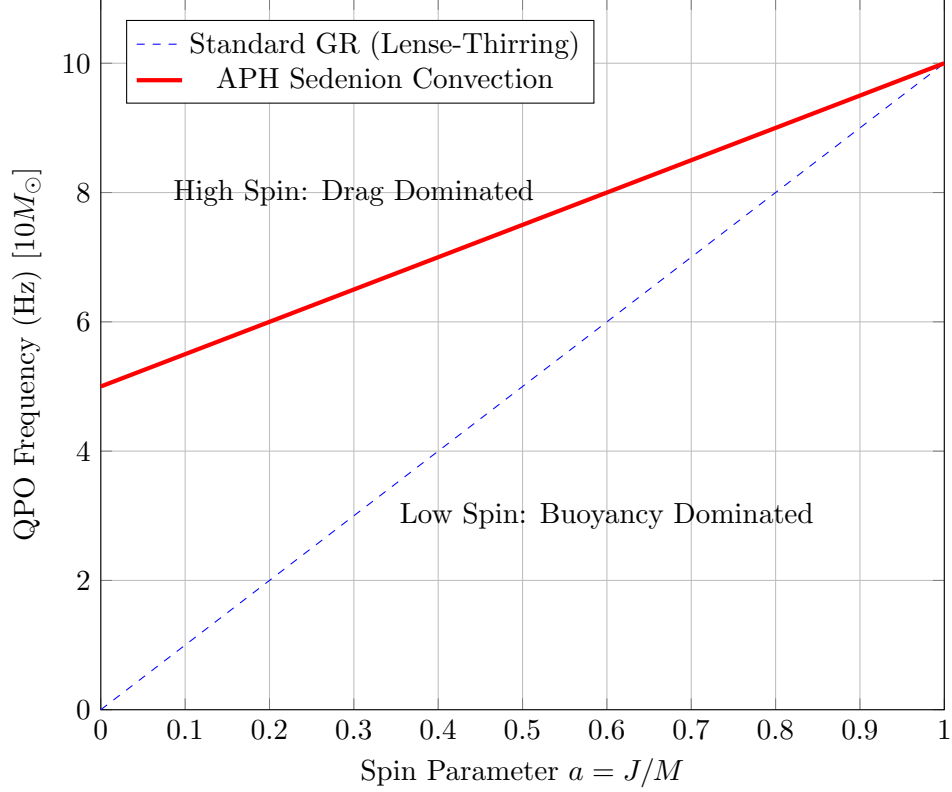


Figure 4: Predicted QPO Spectrum. Standard GR predicts QPO frequency scales linearly with spin (blue dashed). The APH model (red solid) predicts a non-zero intercept at $a = 0$ due to intrinsic geometric buoyancy and a modified slope due to core stiffness.

4.1 The Domain Wall Tension

The Event Horizon is the phase boundary between $\mathbb{O}$ and $\mathbb{S}$. This boundary possesses an intrinsic surface tension σ_{wall} due to the mismatch in topological invariants (Betti numbers) between the G_2 (defect) and S^3 (vacuum) manifolds.

4.2 The Sublimation Rate

The Sedenion core is *hotter* (entropically higher) than the vacuum. The decay rate is governed by the probability of a boundary element *snapping* from $\mathbb{S} \rightarrow \mathbb{O}$. Solving for the equilibrium condition yields an effective temperature:

$$T_{APH} = \frac{\hbar c^3}{8\pi G M} \left(\frac{\sigma_{wall}}{\sigma_{Planck}} \right) \quad (5)$$

This recovers the $1/M$ scaling of Hawking Radiation but identifies the proportionality constant with the *Material Stiffness* of the G_2 manifold.

5 Conclusion

We have constructed a complete thermodynamic life-cycle for gravitational collapse in the APH framework.

1. **Formation:** Sedenion Stars form when $\Gamma \rightarrow \pi$.
2. **Stability:** The core is supported by geometric stiffness and stabilized against singularity formation by Rice-style flux convection.

3. **Dynamics:** The core exhibits convective turnover, generating observable QPOs driven by the algebraic stiffness gradient.
4. **Evaporation:** The object decays via algebraic sublimation of the domain wall.

This model provides a unitary, finite, and geometrically consistent description of black holes, testable via the Gravitational Wave Echo signatures derived in [1] and X-ray timing analyses.

References

- [1] Schutz, A. M. (2025). *Gravitational Wave Echoes from the Algebraic Domain Wall*.
- [2] Schutz, A. M. (2025). *Pregeometric Cosmology: The Sedenion Scream*.
- [3] Reggiani, S. (2024). *The Geometry of Sedenion Zero Divisors*. arXiv:2411.18881.